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Full study note like chapter 2

ChatGPT gpt-6

Chapter 3: Data Preprocessing

Complete Study Notes | Data Mining: Concepts and Techniques, Han, Kamber & Pei, 3rd Edition

These notes combine your **Chapter 3 Lecture Notes** and **Prof. Dr. M. Shamim Kaiser's Data Preprocessing slides**. I've organized them as a chapter-level study guide, with explanations, formulas, worked numerical examples, practical scenarios, exam tips, and practice questions.

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1. Introduction to Data Preprocessing

1.1 What is data preprocessing?

Data preprocessing is the process of converting raw data into a clean, consistent, reduced, and suitable format before applying data mining algorithms.

Real-world data is rarely perfect. It may contain:

- Missing values
- Incorrect values
- Duplicate records
- Inconsistent formats
- Noise and outliers
- Redundant attributes
- Different measurement units

For example, imagine a bank collecting customer information from three branches.

Customer	Age	Salary	City
A	25	40,000	Dhaka
B	NULL	55,000	Dhanmondi
C	999	60,000	DHAKA
A	25	40,000	Dhaka

Problems:

- Customer B has a missing age.
- Customer C has an unrealistic age.
- Dhaka and DHAKA use inconsistent capitalization.
- Customer A appears twice.

If we directly apply a machine learning algorithm to this dataset, the results may be inaccurate.

This leads to an important principle:

Garbage In, Garbage Out

If poor-quality data enters a data mining system, the output will also likely be poor-quality.

1.2 Why is preprocessing important?

According to the lecture notes, preprocessing can consume approximately 60–80% of the effort in a data mining project.

It is important because it:

1. Improves the accuracy of mining results.
2. Handles missing and incorrect values.
3. Removes unnecessary or redundant information.
4. Makes data consistent across multiple sources.
5. Reduces storage and computational requirements.
6. Makes data easier to understand and interpret.

1.3 Six dimensions of data quality

Data quality has six major dimensions.

Dimension	Meaning	Example
Accuracy	Values are correct	Age recorded as 999
Completeness	Required values are present	Missing customer phone number
Consistency	Data agrees across systems	Different birthdates for the same customer
Timeliness	Data is up to date	Old address after a customer moves
Believability	Data can be trusted	Implausible salary figures
Interpretability	Data is understandable	Abbreviations without explanations

Exam question: Explain the six dimensions of data quality with examples.

Remember: Accuracy, Completeness, Consistency, Timeliness, Believability, Interpretability.

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1.4 Sources of dirty data

A. Incomplete data

Data is incomplete when some attribute values are unavailable.

Possible causes:

- Information was not collected.
- Some attributes were considered unimportant.
- Users did not provide information.
- Data was lost during collection or storage.

Example: A customer record contains a name and address but no phone number.

B. Noisy data

Noise means random errors or unwanted variations in data.

Possible causes:

- Faulty data collection instruments.
- Human data-entry errors.
- Computer errors.
- Transmission errors.

Example: A temperature sensor records 250°C instead of 25°C.

C. Inconsistent data

Data is inconsistent when different records or sources use conflicting formats, codes, or values.

Examples:

- `Male/Female` versus `M/F`
- `Dhaka` versus `DHAKA`
- `01-Jan-2010` versus `1/1/10`
- Weight recorded in kilograms in one database and pounds in another.

1.5 Four major data preprocessing tasks

1. Data Cleaning

Fill missing values, smooth noise, identify outliers, and resolve inconsistencies.

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2. Data Integration

Combine data from multiple sources into a coherent dataset.

3. Data Reduction

Reduce the volume or dimensionality of data while preserving useful analytical results.

4. Data Transformation

Convert data into a suitable form using normalization, aggregation, discretization, and other techniques.

These tasks are not necessarily performed in a strict sequence. They can interact and repeat.

For example, integration may reveal duplicate attributes, which can then be removed during data reduction.

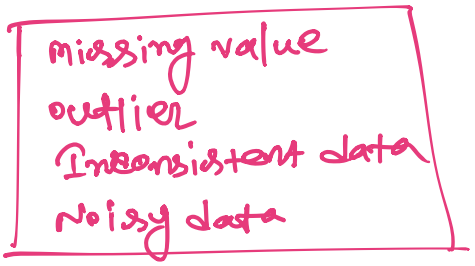
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2. Data Cleaning

2.1 What is data cleaning?

Data cleaning, also known as data cleansing, is the process of:

- Filling missing values.
- Smoothing noisy data.
- Identifying and handling outliers.
- Resolving inconsistent data.



It is often one of the most labor-intensive stages of preprocessing.

2.2 Handling missing values

A missing value occurs when an attribute has no recorded value for a particular tuple.

Example:

Student	Age	Marks
A	22	85
B	NULL	75
C	24	90
D	23	NULL

There are several ways to handle missing values.

Method 1: Ignore the tuple

Remove the entire record containing missing data.

Example:

Student	Age	Marks
A	22	85
C	24	90

When to use:

- When the missing information makes the record unusable.
- The lecture notes particularly mention a missing class label in classification.

Disadvantage: You lose potentially useful information.

Method 2: Fill in manually

A domain expert manually enters the missing value.

Example:

A university administrator checks the original admission form and enters the missing age.

When to use: Small datasets where manual verification is practical.

Disadvantage: Time-consuming and impractical for large datasets.

Method 3: Use a global constant

Replace missing values with a fixed value.

Examples:

- `Unknown`
- `Not Available`
- `0`
- `-1`

Example:

Student	Age
A	22
B	Unknown

Disadvantage: An arbitrary value may confuse the mining algorithm.

For example, replacing missing age with zero could make the algorithm interpret that the student is actually zero years old.

Method 4: Use the attribute mean

Replace a missing numerical value with the mean of that attribute.

Formula:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

Example:

Ages:

$$20,\ 22,\ 24,\ 26,\ NULL$$

Calculate the mean:

$$\bar{x} = \frac{20 + 22 + 24 + 26}{4}$$

$$\bar{x} = \frac{92}{4} = 23$$

Therefore:

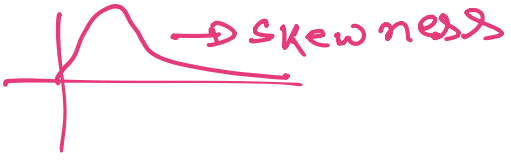
$$NULL \rightarrow 23$$

Updated dataset:

$$20,\ 22,\ 24,\ 26,\ 23$$

When to use: Continuous numerical attributes with roughly symmetrical distributions.

Disadvantage: The mean can be affected by extreme values.



Method 5: Use the class-conditional mean

Instead of calculating the mean of the entire dataset, calculate the mean within the same class.

Example:

Student	Result Category	Marks
A	Pass	70
B	Pass	80
C	Pass	NULL
D	Fail	30
E	Fail	40

For student C, calculate the mean of the Pass category:

$$\bar{x}_{Pass} = \frac{70 + 80}{2} = 75$$

Therefore:

$$NULL \rightarrow 75$$

Why is this better?

The overall mean would be:

$$\bar{x} = \frac{70 + 80 + 30 + 40}{4} = 55$$

Using 55 for a student in the Pass category would be less representative.

Method 6: Use the most probable value

Estimate the missing value using statistical or predictive methods.

Examples:

- Regression
- Bayesian inference
- Other predictive models

Suppose a student's marks are missing. We can use attendance, assignment marks, previous exam marks, and other information to estimate the most probable final marks.

Important for exams

The lecture notes identify the most probable value approach as the most sophisticated because it uses existing information to estimate the missing value instead of inserting an arbitrary constant.

Comparison of missing-value methods

Method	Main idea	Limitation
Ignore tuple	Delete record	Information loss
Manual filling	Expert enters value	Time-consuming
Global constant	Use fixed value	May introduce bias
Attribute mean	Overall mean	Affected by outliers
Class mean	Mean within same class	Requires class information
Most probable	Predict missing value	Requires a suitable model

2.3 Noisy data

Noise refers to random errors or unwanted variations in measured data.

Example:

Suppose a temperature sensor records:

$$20,\ 21,\ 22,\ 21,\ 23,\ 95,\ 22$$

The value 95 may be an error because the other values are close to 20–23.

The goal of smoothing is to reduce unwanted variation without destroying the important pattern.

The lecture materials discuss three important approaches:

1. Binning
2. Regression
3. Outlier analysis

2.4 Binning

Binning is a data smoothing technique that divides sorted values into groups called bins.

It can use:

1. Equal-width partitioning
2. Equal-frequency partitioning

After dividing values into bins, we can smooth them using:

1. Bin mean
2. Bin median

3. Bin boundaries

A. Equal-frequency binning

Each bin contains approximately the same number of observations.

Consider the sorted dataset:

$$4,\ 8,\ 15,\ 21,\ 21,\ 24,\ 25,\ 28,\ 34$$

We need three bins, each containing three values.

Step 1: Divide into bins

$$B_1 = (4, 8, 15) \rightarrow (9, 9, 9)$$

$$B_2 = (21, 21, 24) \rightarrow (22, 22, 22)$$

$$B_3 = (25, 28, 34) \rightarrow (29, 29, 29)$$

B. Smoothing by bin mean

Replace every value in a bin with that bin's mean.

Formula:

$$Mean = \frac{\sum x_i}{n}$$

Bin 1:

$$\mu_1 = \frac{4 + 8 + 15}{3} = \frac{27}{3} = 9$$

Replace all three values with 9.

$$(9, 9, 9)$$

Bin 2:

$$\frac{21 + 21 + 24}{3} = \frac{66}{3} = 22$$

$$(22, 22, 22)$$

Bin 3:

$$\frac{25+28+34}{3} = \frac{87}{3} = 29$$

$$(29, 29, 29)$$

Final answer: Smoothing by mean

Original bin	Smoothed bin
4, 8, 15	9, 9, 9
21, 21, 24	22, 22, 22
25, 28, 34	29, 29, 29

Remember: Every value in the same bin becomes the bin mean.

C. Smoothing by bin median

Replace all values in a bin with its median.

For an odd number of sorted observations, the median is the middle value.

Bin 1:

$$(4, 8, 15) \rightarrow 8$$

Bin 2:

$$(21, 21, 24) \rightarrow 21$$

Bin 3:

$$(25, 28, 34) \rightarrow 28$$

Final result:

$$8, 8, 8,\ 21, 21, 21,\ 28, 28, 28$$

D. Smoothing by bin boundaries

Replace each value with the nearest boundary of its bin.

The boundaries are the minimum and maximum values in each bin.

Bin 1:

$$(4, 8, 15)$$

Boundaries:

$$4,\ 15$$

- 4 is already a boundary.
- 8 is closer to 4 than 15.
- 15 is already a boundary.

Result:

$$(4, 4, 15)$$

Bin 2:

$$(21, 21, 24)$$

Boundaries:

$$21,\ 24$$

Result:

$$(21, 21, 24)$$

Bin 3:

$$(25, 28, 34)$$

Boundaries:

$$25,\ 34$$

- 25 is a boundary.
- 28 is closer to 25.
- 34 is a boundary.

Result:

$$(25, 25, 34)$$

Complete comparison

Original bin	By mean	By median	By boundaries
4, 8, 15	9, 9, 9	8, 8, 8	4, 4, 15
21, 21, 24	22, 22, 22	21, 21, 21	21, 21, 24
25, 28, 34	29, 29, 29	28, 28, 28	25, 25, 34

Common exam question

Question: Given nine sorted values, divide them into three equal-frequency bins and smooth using mean, median, and boundaries.

You should be able to calculate all three methods independently.

E. Equal-width binning

Instead of giving each bin the same number of values, equal-width binning gives each bin the same numerical range.

Formula:

$$W = \frac{Max - Min}{N}$$

Where:

- W = width of each bin
- Max = maximum value
- Min = minimum value
- N = number of bins

Example:

Dataset:

$$10,\ 20,\ 30,\ 40,\ 50,\ 60$$

Number of bins:

$$N = 3$$

Calculate width:

$$W = \frac{60 - 10}{3}$$

$$W = 16.67$$

Intervals:

- Bin 1: [10, 26.67)
- Bin 2: [26.67, 43.33)
- Bin 3: [43.33, 60]

Values are assigned according to their intervals.

Difference between equal-width and equal-frequency:

Equal-width	Equal-frequency
Same numerical range per bin	Approximately same number of records per bin
Number of values per bin may differ	Numerical widths may differ
Sensitive to outliers	Better balanced record distribution

2.5 Regression for smoothing

Regression models the relationship between variables and uses predicted values to reduce noise.

Simple linear regression:

$$y = wx + b$$

Where:

- y = predicted value
- x = input variable
- w = slope
- b = intercept

Example:

Suppose we want to predict examination marks based on study hours.

Study hours	Actual marks
1	45
2	55
3	65
4	74

Suppose the fitted regression equation is:

$$y = 10x + 35$$

For four hours:

$$y = 10(4) + 35 = 75$$

The predicted value is 75. A noisy observation can be replaced or supplemented by this fitted value.

Types of regression mentioned in the notes:

- Linear regression
- Multiple regression
- Logistic regression

Multiple regression uses several independent variables.

$$y = b_0 + b_1x_1 + b_2x_2 + \cdots + b_nx_n$$

2.6 Outlier analysis

An outlier is a data object that differs significantly from most other observations.

Example:

$$20,\ 22,\ 21,\ 23,\ 22,\ 180$$

The value 180 is a potential outlier.

Methods mentioned in the lecture materials:

- Clustering: points that do not belong to any cluster are candidates for outliers.
- Statistical tests.
- Density-based methods.

Outliers are not always errors. For example, a transaction worth Tk 10 million may be unusual but legitimate.

Therefore, investigate before removing an outlier.

2.7 Inconsistent data

Inconsistencies can occur because of:

A. Format differences

- `01-Jan-2010`
- `1/1/10`
- `January 1, 2010`

Solution: Convert everything to a standard format.

B. Unit differences

- Inches vs centimeters
- Kilograms vs pounds
- USD vs EUR

Solution: Convert values into a common unit.

C. Constraint violations

Examples:

- Negative age
- Impossible birthdate
- Salary below an allowed minimum

Solution: Apply domain constraints and validation rules.

D. Duplicate records

The same customer may appear multiple times with slightly different spellings.

Solution: Use duplicate detection and entity matching.

2.8 Data cleaning as an iterative process

Data cleaning is not a one-time operation.

The lecture notes describe this process:

1. **Discrepancy detection**Identify missing values, outliers, invalid formats, and rule violations.
2. **Data transformation**Correct values, standardize formats, recode attributes, and merge duplicates.
3. **Verification**Check whether the corrections produced the intended result.
4. **Repeat**Continue detecting and correcting problems because one correction may reveal another.

The lecture slides also mention:

- Metadata and domain knowledge
- Data scrubbing tools
- Data auditing tools
- ETL and migration tools

- Potter's Wheel, an interactive data-cleaning system

3. Data Integration

3.1 What is data integration?

Data integration combines data from multiple heterogeneous sources into a single coherent data store.

Sources can include:

- Relational databases
- Flat files
- Data warehouses
- Data cubes
- External systems

Example:

A bank has separate databases for:

- Customer information
- Loan information
- Transaction information

Data integration combines these sources to provide a unified view of each customer.

However, integration creates several problems:

1. Entity identification
2. Redundant data
3. Data value conflicts

3.2 Entity identification problem

The entity identification problem asks:

Do records from different sources refer to the same real-world entity?

Example:

Database A	Database B
J. Smith	John Smith
Customer ID: 101	Customer ID: C-901
Dhaka	DHAKA

These records may represent the same customer even though their names and IDs differ.

Solutions:

- Check unique identifiers.
- Compare descriptions and attributes.
- Use metadata.
- Apply record linkage algorithms.
- Use domain knowledge.

Important: Two attributes with the same name do not necessarily represent the same concept.

For example, `Customer_ID` in one system may be a sequential number, while another system uses a hashed email.

3.3 Redundant data

An attribute is redundant if it can be derived from another attribute or a combination of attributes.

Example:

Height (cm)	Height (m)
170	1.70
180	1.80
160	1.60

Both attributes represent the same information.

Keeping both may be unnecessary.

Redundancy can be detected using correlation analysis.

Two important approaches:

- Chi-squared test for nominal attributes.
- Pearson correlation for numerical attributes.

3.4 Chi-squared test (χ^2)

The chi-squared test is used to examine whether two categorical attributes are independent or associated.

Example:

We want to determine whether a student's programme is associated with their preferred exam mode.

Suppose we have 200 students.

Programme	Online	Offline	Total
CSE	60	40	100
EEE	30	70	100
Total	90	110	200

Observed frequencies:

- CSE and Online = 60
- CSE and Offline = 40
- EEE and Online = 30
- EEE and Offline = 70

Step 1: Understand observed and expected frequencies

- Observed frequency (O): Actual count in the dataset.
- Expected frequency (E): Count expected if the two attributes were independent.

Formula:

$$E_{ij} = \frac{Row\ Total \times Column\ Total}{Grand\ Total}$$

Step 2: Calculate expected frequencies

For CSE and Online:

$$E_{11} = \frac{100 \times 90}{200} = 45$$

For CSE and Offline:

$$E_{12} = \frac{100 \times 110}{200} = 55$$

For EEE and Online:

$$E_{21} = \frac{100 \times 90}{200} = 45$$

For EEE and Offline:

$$E_{22} = \frac{100 \times 110}{200} = 55$$

Expected frequency table:

Programme	Online	Offline
CSE	45	55
EEE	45	55

Step 3: Apply the chi-squared formula

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Calculate each cell:

$$\chi^2 = \frac{(60 - 45)^2}{45} + \frac{(40 - 55)^2}{55}$$

$$+ \frac{(30 - 45)^2}{45} + \frac{(70 - 55)^2}{55}$$

$$= \frac{225}{45} + \frac{225}{55} + \frac{225}{45} + \frac{225}{55}$$

$$= 5 + 4.09 + 5 + 4.09$$

$$\boxed{\chi^2 = 18.18}$$

Step 4: Calculate degrees of freedom

Formula:

$$df = (r - 1)(c - 1)$$

Where:

- r = number of rows
- c = number of columns

Here:

$$df = (2 - 1)(2 - 1) = 1$$

Step 5: Compare with critical value

At significance level:

$$\alpha = 0.001$$

For 1 degree of freedom, the critical value is approximately:

$$\chi^2_{crit.} = 10.83$$

Our calculated value:

$$18.18 > 10.83$$

Therefore, reject the null hypothesis of independence.

Conclusion: Programme and preferred exam mode are associated.

Exam result

$$\chi^2 = 18.18, \quad df = 1$$

How to interpret χ²

Result	Interpretation
Small χ²	Observed values are close to expected values; little evidence against independence
Large χ²	Observed values differ significantly from expected values; evidence of association

Important exam mistake: Chi-squared indicates whether there is evidence of an association. It does not measure the strength of the correlation.

Also, association does not prove causation.

3.5 Pearson correlation coefficient

Pearson correlation measures the direction and strength of a linear relationship between two numerical attributes.

Formula:

$$r_{A,B} = \frac{\sum_{i=1}^{n}(a_i - \bar{A})(b_i - \bar{B})}{n\sigma_A\sigma_B} = \frac{\sum A_i B_i - n\bar{A}\bar{B}}{n\sigma_A\sigma_B}$$

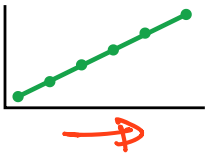
The value of Pearson's correlation coefficient always lies between -1 and +1.

$$-1 \leq r \leq +1$$

Interpretation

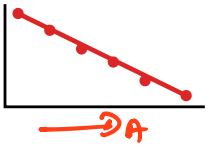
Correlation	Meaning
+1	Perfect positive linear correlation
0	No linear correlation
-1	Perfect negative linear correlation
Close to +1	Strong positive relationship
Close to -1	Strong negative relationship

Understanding the three main cases



Positive correlation

As A increases, B tends to increase.



Negative correlation

As A increases, B tends to decrease.



No clear linear correlation

No consistent straight-line pattern.

Worked example

Suppose:

$$A = (2, 4, 5, 7, 8)$$

$$B = (50, 60, 65, 80, 85)$$

The lecture slides provide the Pearson correlation result:

$$r = \frac{137}{137.56}$$

$$\boxed{r \approx 0.996}$$

This indicates a very strong positive linear relationship between study hours and marks.

Why is Pearson correlation useful in data integration?

Suppose two attributes have a correlation of 0.99.

They may contain highly similar information. We can investigate whether one attribute is redundant.

However, high correlation does not automatically mean one attribute should be deleted. Domain knowledge is also important.

3.6 Covariance

Covariance measures how two numerical variables vary together.

Formula:

$$Cov(A,B) = \frac{1}{n}\sum_{i=1}^{n}(a_i - \bar{A})(b_i - \bar{B})$$

It can also be expressed as:

$$Cov(A,B) = E[AB] - \bar{A}\bar{B}$$

Interpretation:

- Positive covariance: Variables tend to move in the same direction.
- Negative covariance: Variables tend to move in opposite directions.
- Zero covariance: No linear co-variation is indicated.

Worked example from the lecture slides

Two stocks have weekly closing prices:

$$A = (12, 15, 11, 18, 14)$$

$$B = (30, 36, 28, 40, 31)$$

Means:

$$\bar{A} = 14$$

$$\bar{B} = 33$$

Calculate deviations:

Week	A deviation	B deviation	Product
1	-2	-3	6
2	1	3	3
3	-3	-5	15
4	4	7	28
5	0	-2	0

Sum of products:

$$6 + 3 + 15 + 28 + 0 = 52$$

Population covariance:

$$Cov(A, B) = \frac{52}{5}$$

$$\boxed{Cov(A, B) = 10.4}$$

Since covariance is positive, the stocks tend to move in the same direction.

The slides report Pearson correlation of approximately 0.97.

Pearson correlation vs covariance

Pearson correlation	Covariance
Measures linear relationship	Measures joint variation
Always between -1 and +1	No fixed range
Scale-independent	Depends on units
Easier to compare across datasets	Magnitude is harder to interpret

Important concept:

Independence implies zero covariance under the usual finite-moment assumptions, but zero covariance does not generally imply independence.

3.7 Resolving data value conflicts

When integrating different sources, the same attribute may have different representations.

Conflict type	Example	Solution
Unit difference	kg vs lb	Convert to one unit
Scale difference	USD vs EUR	Apply exchange-rate conversion
Encoding difference	M/F vs 1/2	Map to unified codes
Temporal conflict	Different recording dates	Use the appropriate latest value and flag discrepancies

4. Data Reduction

4.1 What is data reduction?

Data reduction creates a smaller representation of a dataset while preserving the same or nearly the same analytical results.

Why do we need it?

Imagine a company has 100 million transaction records and 200 attributes.

Analyzing all the data may require significant memory, storage, and processing time.

Data reduction makes the dataset easier and faster to analyze.

Three major categories:

1. Dimensionality reduction
2. Numerosity reduction
3. Data compression

4.2 Difference between the three categories

Category	What is reduced?	Example
Dimensionality reduction	Number of attributes	200 features → 20 features
Numerosity reduction	Number of data objects or records	100 million records → representative sample
Data compression	Storage representation	Compress a large dataset into a smaller encoding

This distinction is extremely important for exams.

4.3 Dimensionality reduction

Dimensionality reduction reduces the number of attributes or features.

Example:

A student dataset contains:

- Age
- Attendance
- Assignment marks
- Quiz marks
- Midterm marks
- Final marks
- Total marks
- Grade

Some attributes may contain overlapping information.

We can reduce the number of attributes while preserving useful information.

The lecture materials cover:

1. Principal Component Analysis (PCA)
2. Attribute subset selection
3. Wavelet transforms

A. Principal Component Analysis (PCA)

PCA transforms the original numerical attributes into a smaller number of new, uncorrelated variables called principal components.

These components capture as much of the original data's variance as possible.

Example:

Suppose a dataset has 10 numerical attributes.

After applying PCA, we may find that the first three principal components capture most of the important variation.

We can represent the dataset using three components instead of ten.

$$10\ dimensions \rightarrow 3\ dimensions$$

Important PCA terms

Principal component: A new axis or direction representing a combination of original attributes.

Eigenvector: Represents the direction of a principal component.

Eigenvalue: Represents the amount of variance captured by its corresponding eigenvector.

PCA procedure

1. **Normalize the input data** Put attributes on comparable scales so that large numerical ranges do not dominate.
2. **Compute the covariance matrix** Identify relationships among the numerical attributes.
3. **Calculate eigenvalues and eigenvectors** Eigenvectors provide directions; eigenvalues indicate the variance captured.
4. **Sort eigenvectors** Sort them according to decreasing eigenvalues.
5. **Select top-k eigenvectors** Keep the components that capture the most variance.
6. **Project the original data** Transform the original data into the reduced-dimensional space.

PCA sequence to memorize

$$Normalize \rightarrow Covariance\ Matrix \rightarrow Eigenvectors/Eigenvalues \rightarrow Sort \rightarrow Project$$

Important PCA properties:

- PCA is an unsupervised technique.
- It does not use class labels.
- It works with numerical attributes.
- It is most useful when attributes are correlated.
- It can reduce computational complexity.

B. Attribute subset selection

Attribute subset selection selects a smaller set of useful attributes from the original dataset.

Unlike PCA, it selects original attributes rather than creating new combined attributes.

Suppose:

$$A = \{A_1, A_2, A_3, A_4, A_5, A_6\}$$

After feature selection:

$$A' = \{A_1, A_4, A_6\}$$

The goal is to preserve important information while removing unnecessary features.

Four approaches from the lecture materials:

1. Stepwise forward selection

- Start with an empty set.
- Add the best attribute at each step.
- Continue until the stopping condition is reached.

Example:

$$\{\} \rightarrow \{A_1\} \rightarrow \{A_1, A_4\} \rightarrow \{A_1, A_4, A_6\}$$

2. Stepwise backward elimination

- Start with all attributes.
- Remove the least useful attribute at each step.

3. Forward-backward combination

- Add the best attributes.
- Remove the least useful attributes.
- Repeat.

4. Decision tree induction

Attributes not appearing in the decision tree may be considered unnecessary for that tree's predictions.

C. Wavelet transforms

Wavelet transforms decompose numerical data into different frequency components.

They help represent data using a smaller number of important coefficients.

Applications:

- Signal processing
- Image compression
- Noise removal

The lecture slides mention JPEG 2000 as an example of wavelet-based image compression.

Haar wavelet worked example

Given:

$$X = (2, 2, 0, 2, 3, 5, 4, 4)$$

The slides use:

$$Average = \frac{a + b}{2}$$

$$Detail = \frac{a - b}{2}$$

Level 1: Pair the values

Pairs:

$$(2, 2), (0, 2), (3, 5), (4, 4)$$

Averages:

$$(2, 1, 4, 4)$$

Details:

$$(0, -1, -1, 0)$$

Level 2: Apply the process to averages

$$(2, 1), (4, 4)$$

Averages:

$$(1.5, 4)$$

Details:

$$(0.5, 0)$$

Level 3: Apply the process again

$$(1.5, 4)$$

Average:

$$\frac{1.5 + 4}{2} = 2.75$$

Detail:

$$\frac{1.5 - 4}{2} = -1.25$$

Final coefficient vector:

$$\boxed{X' = (2.75, -1.25, 0.5, 0, 0, -1, -1, 0)}$$

The important idea is that we can keep the strongest coefficients and set weaker coefficients to zero to obtain a smaller representation.

4.4 Numerosity reduction

Numerosity reduction decreases the number of data objects or records while retaining a useful representation of the original data.

The lecture materials discuss:

1. Regression and log-linear models
2. Histograms
3. Clustering
4. Sampling
5. Data cube aggregation

A. Regression models

Instead of storing all data points, store a mathematical model that approximates them.

Simple linear regression:

$$y = wx + b$$

Suppose a dataset has 1 million points that follow a nearly linear pattern.

Instead of storing every point, we may store only:

- Slope w
- Intercept b

The model can estimate the values.

Some poorly fitted outliers may also need to be stored separately.

B. Multiple regression

Uses multiple independent variables:

$$y = b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_n x_n$$

C. Log-linear models

Log-linear models approximate discrete multidimensional probability distributions.

They can estimate a multidimensional distribution using lower-dimensional combinations.

The lecture notes mention their use for categorical variables.

D. Histograms

A histogram divides attribute values into buckets and stores summary information, such as the count or sum for each bucket.

Example:

Suppose sales values are:

$$10, 12, 14, 20, 22, 24, 30, 32, 34$$

Using three equal-width buckets:

Bucket	Range	Values	Count
1	10–18	10, 12, 14	3
2	19–27	20, 22, 24	3
3	28–36	30, 32, 34	3

Instead of retaining every original value for certain analyses, we can store bucket summaries.

Four histogram partitioning methods:

Method	Explanation
Equal-width	All buckets cover the same numerical range
Equal-frequency	Each bucket contains approximately the same number of values
V-Optimal	Minimizes weighted variance within buckets
MaxDiff	Places boundaries where consecutive values differ the most

E. Clustering

Clustering groups similar data points together.

Instead of storing every data point, store a representative for each cluster.

Examples of cluster representatives:

- Centroid

- Medoid

Suppose 10,000 customer records form five natural groups.

We may represent the groups using five cluster representatives for certain analytical tasks.

Clustering works best when data has natural groups. It is less effective when data is widely scattered or has no clear grouping.

F. Sampling

Sampling represents a large dataset using a smaller subset.

Let:

- N = total population size
- s = sample size

Usually:

$$s \ll N$$

Example:

A bank has 1,000,000 transactions. Instead of analyzing all transactions during an initial exploratory task, it selects 10,000 representative transactions.

The lecture slides discuss four methods.

1. Simple Random Sampling Without Replacement (SRSWOR)

Every object has an equal chance of being selected, and once selected, it is not returned.

Example:

Select 10 students from a class of 100 without selecting the same student twice.

2. Simple Random Sampling With Replacement (SRSWR)

Every selected object is returned to the population before the next draw.

An object can be selected more than once.

3. Cluster Sampling

Divide the population into clusters and randomly select clusters.

Example:

A university has 50 classrooms. Randomly select five classrooms and study students from those classrooms.

4. Stratified Sampling

Divide the population into strata and randomly sample from each stratum.

Example:

A university has:

Department	Students
CSE	500
EEE	300
BBA	200
Total	1,000

Suppose we need a sample of 100 students.

Proportional stratified sampling:

CSE:

$$\frac{500}{1000} \times 100 = 50$$

EEE:

$$\frac{300}{1000} \times 100 = 30$$

BBA:

$$\frac{200}{1000} \times 100 = 20$$

Final sample:

Department	Sample
CSE	50
EEE	30
BBA	20
Total	100

Very important exam concept

Stratified sampling is preferred for highly skewed or imbalanced class distributions because it ensures that minority groups are represented.

Sampling methods comparison

Method	Replacement?	How selection works
SRSWOR	No	Random individual selection
SRSWR	Yes	Random selection with return
Cluster	Not the defining feature	Randomly select clusters
Stratified	Usually without replacement within strata	Sample from every stratum

G. Data cube aggregation

Data cube aggregation summarizes data at a higher level of granularity.

Example:

A company has daily sales records.

If the business wants annual sales, it may aggregate daily sales into monthly, quarterly, or annual totals.

The lecture slides provide this example:

Quarter	Sales (Tk lakh)
Q1	224
Q2	408
Q3	350
Q4	586
Annual	1,568

Calculation:

$$224 + 408 + 350 + 586 = 1568$$

Instead of querying every daily transaction, the system can use the annual aggregate when only annual sales are required.

Data cube terminology:

- Base cuboid: Lowest level of detail.
- Apex cuboid: Highest level of aggregation.

Use the smallest cuboid that contains the information required for the query.

4.5 Data compression

Data compression reduces the storage space needed to represent data.

Two major types:

A. Lossless compression

The original data can be reconstructed exactly.

Examples:

- String compression
- ZIP-style compression

B. Lossy compression

The original data cannot be reconstructed exactly. Only an approximation is available.

Examples:

- JPEG
- Wavelet-based compression

In data mining, lossy compression may be acceptable if the approximation error is sufficiently small for the task.

Lossless vs lossy

Lossless	Lossy
Exact reconstruction	Approximate reconstruction
No information is lost	Some information is discarded
Useful when exact data is required	Useful when small approximation errors are acceptable

---

5. Data Transformation and Discretization

5.1 What is data transformation?

Data transformation converts data into a format that is more suitable for mining algorithms.

For example, a machine learning algorithm may perform poorly if one attribute ranges from 0 to 1 and another ranges from 0 to 1,000,000.

Transformation helps make the data more consistent and useful.

The six transformation strategies covered in the lecture slides are:

1. Smoothing
2. Attribute construction
3. Aggregation
4. Normalization
5. Discretization
6. Concept hierarchy generation

5.2 Smoothing

Smoothing removes noise from data.

Techniques:

- Binning
- Regression
- Clustering

We already covered these techniques in data cleaning.

5.3 Attribute construction

Attribute construction creates new attributes from existing attributes.

Example:

Suppose a dataset contains:

- Height
- Width

We can construct a new attribute:

$$Area = Height \times Width$$

Another example:

From date of birth, we can calculate:

$$Age = Current\ Year - Birth\ Year$$

Why use attribute construction?

A new attribute may provide information that helps a mining algorithm identify patterns more effectively.

5.4 Aggregation

Aggregation combines multiple values into a summarized value.

Example:

Daily sales:

$$100,\ 150,\ 200,\ 250$$

Monthly total:

$$100 + 150 + 200 + 250 = 700$$

Daily data has been converted into a monthly summary.

Aggregation is commonly used in data warehouses and data cubes.

---

6. Normalization

6.1 What is normalization?

Normalization scales numerical attributes into a smaller or specified range.

It is particularly important for distance-based algorithms such as:

- K-Nearest Neighbors (k-NN)
- Clustering
- Support Vector Machines (SVM)

It can also help neural network training converge.

Why is normalization necessary?

Suppose we have two attributes:

Student	Age	Annual income
A	20	200,000

B	25	500,000
C	30	800,000

Age ranges from 20 to 30.

Income ranges from 200,000 to 800,000.

If an algorithm calculates distances directly, income may dominate because of its much larger numerical scale.

Normalization makes the attributes comparable.

The lecture materials cover three methods:

1. Min-max normalization
2. Z-score normalization
3. Decimal scaling

6.2 Min-max normalization

Min-max normalization transforms values into a specified range.

Formula:

$$v' = \frac{v - Min_A}{Max_A - Min_A}(NewMax_A - NewMin_A) + NewMin_A$$

Where:

- v = original value
- Min_A = minimum original value
- Max_A = maximum original value
- $NewMin_A$ = desired minimum
- $NewMax_A$ = desired maximum
- v' = normalized value

Worked example

Given:

$$Min = 10$$

$$Max = 50$$

Normalize:

$$v = 30$$

Target range:

$$[0, 1]$$

Step 1: Substitute into the formula.

$$v' = \frac{30 - 10}{50 - 10}(1 - 0) + 0$$

Step 2: Calculate.

$$v' = \frac{20}{40}$$

$$\boxed{v' = 0.5}$$

The normalized value is 0.5.

Another example: Normalize to [-1, 1]

Given:

$$Min = 10, \quad Max = 50, \quad v = 30$$

Target range:

$$[-1, 1]$$

$$v' = \frac{30 - 10}{50 - 10}(1 - (-1)) + (-1)$$

$$= \frac{20}{40} \times 2 - 1$$

$$= 1 - 1$$

$$\boxed{v' = 0}$$

Min-max normalization properties

- Maps data to a specified range.
- Preserves the relative ordering of values.
- Sensitive to outliers.
- Future values outside the original minimum and maximum may fall outside the target range.

For future data, store the original normalization parameters and apply them consistently.

6.3 Z-score normalization

Z-score normalization transforms data based on the mean and standard deviation.

Formula:

$$v' = \frac{v - \bar{A}}{\sigma_A}$$

Where:

- v = original value
- $\bar{A}$ = mean
- σ_A = standard deviation

The result tells us how many standard deviations a value is from the mean.

Worked example

Given:

$$v = 70$$

$$Mean = 50$$

$$Standard\ Deviation = 10$$

Apply the formula:

$$v' = \frac{70 - 50}{10}$$

$$= \frac{20}{10}$$

$$\boxed{v' = 2}$$

Interpretation:

The original value is two standard deviations above the mean.

Another example

Given:

$$v = 40, \quad Mean = 50, \quad SD = 10$$

$$v' = \frac{40 - 50}{10}$$

$$\boxed{v' = -1}$$

Interpretation: The value is one standard deviation below the mean.

Properties

- Centers data around zero.
- Uses the mean and standard deviation.
- Does not have a fixed output range.
- Useful when minimum and maximum values are unknown.
- Less sensitive to outliers than min-max in the sense that it does not force extreme values to the endpoints, although the mean and standard deviation themselves can still be affected by outliers.

The slides also mention using mean absolute deviation for a more robust scale estimate.

6.4 Decimal scaling normalization

Decimal scaling normalizes values by moving the decimal point.

Formula:

$$v' = \frac{v}{10^j}$$

Where j is the smallest integer such that:

$$\max |v'| < 1$$

Worked example

Dataset:

$$-850,\ 450,\ 920$$

Maximum absolute value:

$$Max|v| = 920$$

We need the smallest integer j such that:

$$\frac{920}{10^j} < 1$$

Try:

$$j = 1 \Rightarrow \frac{920}{10} = 92$$

Not valid.

$$j = 2 \Rightarrow \frac{920}{100} = 9.2$$

Not valid.

$$j = 3 \Rightarrow \frac{920}{1000} = 0.92$$

Valid.

Therefore:

$$j = 3$$

Normalize all values:

$$-850 \rightarrow -0.850$$

$$450 \rightarrow 0.450$$

$$920 \rightarrow 0.920$$

Final answer:

$$\boxed{-0.850,\ 0.450,\ 0.920}$$

6.5 Comparison of normalization techniques

Feature	Min-max	Z-score	Decimal scaling
Formula uses	Min and max	Mean and SD	Maximum absolute value
Output range	Specified range	Unbounded	Usually between -1 and 1
Sensitive to outliers	High	Can be affected	Relatively low
Main idea	Scale to a range	Standardize around mean	Move decimal point

Quick formula sheet

Min-max

$$v' = \frac{v - Min}{Max - Min}$$

Z-score

$$v' = \frac{v - Mean}{SD}$$

Decimal scaling

$$v' = \frac{v}{10^j}$$

7. Discretization

7.1 What is discretization?

Discretization converts continuous numerical values into a limited number of intervals or categories.

Example:

Original ages:

$$12,\ 18,\ 23,\ 35,\ 48,\ 65$$

After discretization:

Age	Category
12	Child
18	Youth
23	Youth
35	Adult
48	Adult
65	Senior

Instead of processing every unique numerical value, a mining algorithm can work with categories.

7.2 Why do we need discretization?

1. Reduces the number of distinct values.
2. Reduces storage and computational requirements.
3. Makes data easier to interpret.
4. Supports algorithms that require categorical attributes.
5. Helps concept hierarchy generation.

7.3 Types of discretization

Discretization can be classified according to three important dimensions.

A. Supervised vs unsupervised

Supervised	Unsupervised
Uses class labels	Does not use class labels
Considers class information	Considers attribute values and distributions
Example: entropy-based	Example: equal-width, equal-frequency

B. Top-down vs bottom-up

Top-down: Start with a broad interval and repeatedly split it into smaller intervals.

Bottom-up: Start with smaller intervals and merge adjacent intervals.

C. Numerical discretization methods

The lecture slides mention:

- Binning
- Histogram analysis

- Cluster analysis
- Decision tree / entropy-based splitting
- ChiMerge

7.4 Equal-width discretization

Equal-width discretization divides the range of values into intervals of equal width.

Formula:

$$W = \frac{Max - Min}{N}$$

Where N is the number of intervals.

Worked example

Dataset:

$$10,\ 20,\ 30,\ 40,\ 50,\ 60$$

Number of intervals:

$$N = 3$$

Calculate width:

$$W = \frac{60 - 10}{3}$$

$$W = 16.67$$

Intervals:

- Interval 1: [10, 26.67)
- Interval 2: [26.67, 43.33)
- Interval 3: [43.33, 60]

Assign values:

Interval	Values
1	10, 20
2	30, 40

3	50, 60

Every interval has approximately the same numerical width.

Disadvantage: Outliers can make the intervals too wide and leave some intervals nearly empty.

7.5 Equal-frequency discretization

Equal-frequency discretization divides the dataset so that each interval contains approximately the same number of values.

Example:

$$10,\ 15,\ 20,\ 25,\ 30,\ 35,\ 40,\ 45,\ 50$$

Number of bins:

$$N = 3$$

Each bin contains three values.

Bin	Values
1	10, 15, 20
2	25, 30, 35
3	40, 45, 50

Unlike equal-width, the numerical width of each bin does not have to be the same.

7.6 Entropy-based discretization

Entropy-based discretization is a supervised method that uses class labels to identify meaningful split points.

It is associated with Fayyad and Irani (1993) in the lecture notes.

The goal is to divide data into intervals with lower class entropy.

What is entropy?

Entropy measures impurity or uncertainty in a dataset.

Formula:

$$Entropy(S) = -\sum_{i=1}^{m} p_i \log_2(p_i)$$

Where:

- S = dataset
- m = number of classes
- p_i = proportion of observations belonging to class i

Entropy examples

Case 1: Completely pure dataset

Suppose a dataset contains:

- 10 Pass
- 0 Fail

$$p_{Pass} = 1, \quad p_{Fail} = 0$$

$$Entropy(S) = -(1 \log_2 1 + 0 \log_2 0)$$

$$\boxed{Entropy(S) = 0}$$

There is no uncertainty.

Case 2: Equally distributed classes

Suppose:

- 5 Pass
- 5 Fail

$$p_{Pass} = 0.5, \quad p_{Fail} = 0.5$$

$$Entropy(S) = -(0.5 \log_2 0.5 + 0.5 \log_2 0.5)$$

$$= -(0.5(-1) + 0.5(-1))$$

$$\boxed{Entropy(S) = 1}$$

There is maximum uncertainty for two equally represented classes.

Information gain

Suppose a candidate split point T divides dataset S into two subsets:

- S_1
- S_2

Information gain:

$$Gain(S,T) = Entropy(S) - \frac{|S_1|}{|S|}Entropy(S_1) - \frac{|S_2|}{|S|}Entropy(S_2)$$

Where:

- $|S|$ = total number of records
- $|S_1|$ = records in first subset
- $|S_2|$ = records in second subset

The algorithm chooses the split with the highest information gain.

It repeats the process recursively.

The MDL (Minimum Description Length) criterion is used as a stopping rule to avoid unnecessary splits.

General procedure

1. Sort the numerical values.
2. Consider possible split points.
3. Calculate entropy for the original dataset.
4. Calculate the weighted entropy of each split.
5. Calculate information gain.
6. Select the split with maximum gain.
7. Repeat recursively.
8. Stop when the MDL criterion says further splitting is not justified.

7.7 ChiMerge discretization

ChiMerge is a supervised, bottom-up discretization method.

It begins with small intervals and merges adjacent intervals based on their class distributions.

The method uses the chi-squared statistic.

General process:

1. Create initial intervals.

2. Calculate chi-squared values between adjacent intervals.
3. Find adjacent intervals with the lowest chi-squared value.
4. Merge them.
5. Repeat until a stopping condition is reached.

The lecture slides describe ChiMerge as merging adjacent intervals with the lowest χ^2.

7.8 The 3-4-5 rule

The 3-4-5 rule divides numerical data into approximately equal-width intervals based on the most significant digit.

It chooses 3, 4, or 5 intervals to produce human-readable ranges.

The objective is to make numerical intervals easier to interpret.

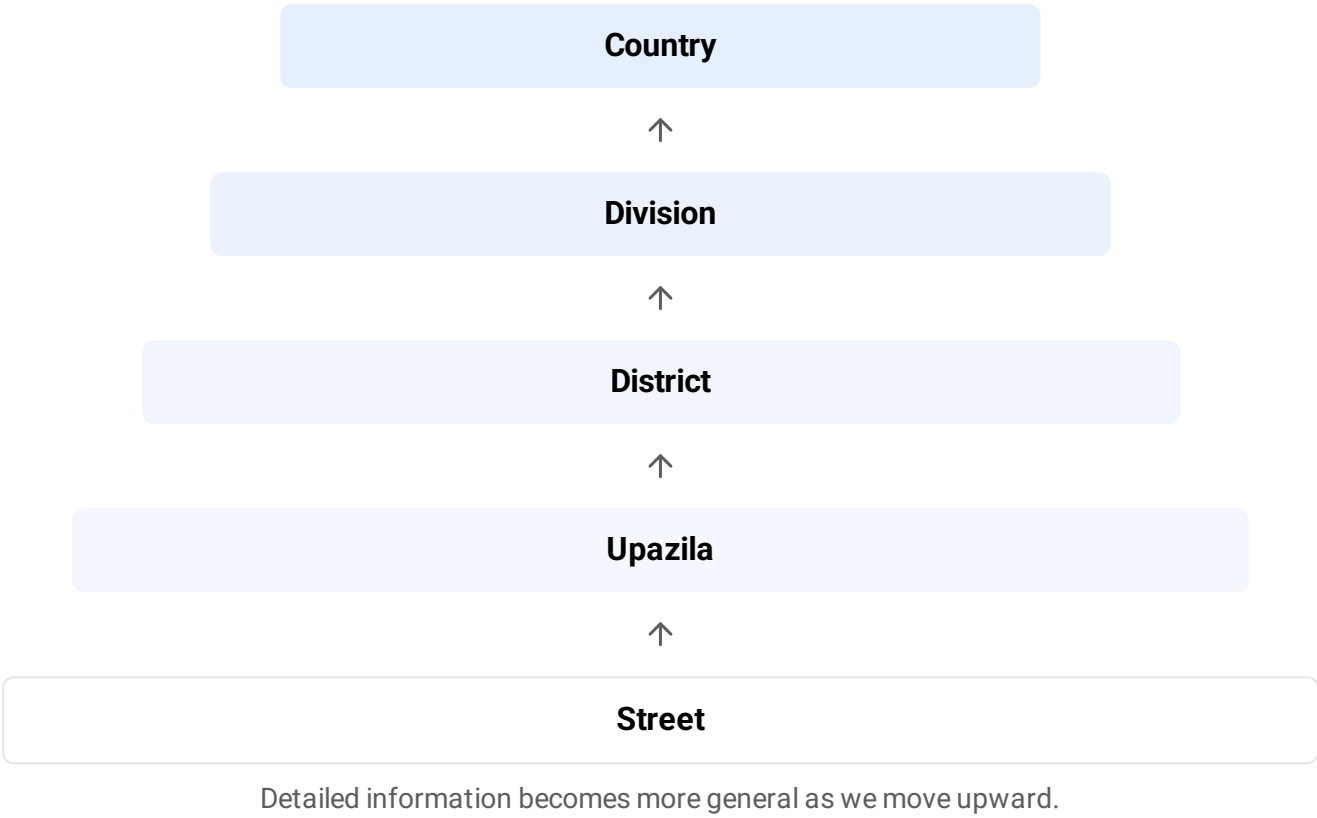
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8. Concept Hierarchy Generation

8.1 What is a concept hierarchy?

A concept hierarchy organizes detailed values into higher-level concepts.

Example:



Detailed information becomes more general as we move upward.

Example:

$$Street \rightarrow City \rightarrow District \rightarrow Division \rightarrow Country$$

This allows data to be analyzed at different levels of detail.

For example:

- Street-level sales
- City-level sales
- District-level sales
- Country-level sales

8.2 Methods of concept hierarchy generation

Method 1: Domain expert specification

A domain expert defines the hierarchy directly.

Example:

$$Street < City < District < Division < Country$$

Method 2: Explicit data grouping

A user manually groups values into higher-level concepts.

Example:

$$\{Savar, Dhamrai\} \subset Dhaka\ District$$

Method 3: Number of distinct values

Attributes with fewer distinct values are placed higher in the hierarchy.

Example from the lecture slides:

Attribute	Approximate number of distinct values
Country	1
Division	8
District	64
Upazila	495
Street	Many

The hierarchy moves from fewer distinct values to more detailed values.

Limitation: This heuristic can fail when the number of distinct values does not reflect the actual semantic relationship.

Method 4: Information gain

Attributes with higher information gain are placed lower in the hierarchy, closer to the detailed values or leaves.

The goal is to preserve meaningful information in the hierarchy.

---

9. Important Comparison Tables

9.1 Data cleaning vs data integration vs data reduction vs transformation

Task	Main purpose	Examples
Cleaning	Improve data quality	Missing value handling, binning
Integration	Combine different sources	Entity matching, conflict resolution
Reduction	Create smaller representation	PCA, sampling, clustering
Transformation	Change representation	Normalization, discretization

9.2 Dimensionality reduction vs numerosity reduction

Dimensionality reduction	Numerosity reduction
Reduces attributes	Reduces data objects or volume
Focuses on features	Focuses on records or representations
PCA	Sampling
Attribute selection	Histograms
Wavelet transforms	Clustering
Example: 100 features to 10	Example: 1 million records to 10,000

9.3 Supervised vs unsupervised discretization

Supervised	Unsupervised
Uses class labels	Does not use class labels
Entropy-based	Equal-width

Decision tree splitting	Equal-frequency
ChiMerge	Histogram analysis

9.4 Top-down vs bottom-up discretization

Top-down	Bottom-up
Start with a large interval	Start with smaller intervals
Split intervals	Merge intervals
Decision tree / entropy splitting	ChiMerge

9.5 Lossless vs lossy compression

Lossless	Lossy
Exact reconstruction	Approximate reconstruction
No information loss	Some information loss
Exact records needed	Small approximation error acceptable

---

10. Important Formulas for the Exam

1. Mean

$$\bar{x} = \frac{\sum x_i}{n}$$

2. Equal-width binning

$$W = \frac{Max - Min}{N}$$

3. Chi-squared test

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

4. Expected frequency

$$E = \frac{Row\ Total \times Column\ Total}{Grand\ Total}$$

5. Chi-squared degrees of freedom

$$df = (r-1)(c-1)$$

6. Pearson correlation

$$r_{A,B} = \frac{\sum(a_i - \bar{A})(b_i - \bar{B})}{n\sigma_A\sigma_B}$$

7. Covariance

$$Cov(A,B) = \frac{1}{n}\sum(a_i - \bar{A})(b_i - \bar{B})$$

8. Linear regression

$$y = wx + b$$

9. Multiple regression

$$y = b_0 + b_1x_1 + \cdots + b_nx_n$$

10. Min-max normalization

$$v' = \frac{v - Min}{Max - Min}(NewMax - NewMin) + NewMin$$

11. Z-score normalization

$$v' = \frac{v - \bar{A}}{\sigma_A}$$

12. Decimal scaling

$$v' = \frac{v}{10^j}$$

13. Entropy

$$Entropy(S) = -\sum p_i \log_2 p_i$$

14. Information gain

$$Gain(S,T) = Entropy(S) - \frac{|S_1|}{|S|}Entropy(S_1) - \frac{|S_2|}{|S|}Entropy(S_2)$$

11. Common Exam Mistakes

Pay special attention to these points.

1. **Confusing equal-width and equal-frequency binning.** Equal-width means equal numerical range; equal-frequency means approximately equal records.

2. **Confusing bin mean with bin median.** Calculate the representative value separately for each bin.

3. **Using the wrong chi-squared expected frequency.** Always calculate expected values using row total × column total / grand total.

4. **Thinking chi-squared measures correlation strength.** It tests evidence of association, not the strength of correlation.

5. **Confusing Pearson correlation and covariance.** Pearson correlation is scale-independent; covariance depends on units.

6. **Confusing dimensionality reduction and numerosity reduction.** One reduces attributes; the other reduces records or their representation.

7. **Forgetting PCA's order.** Normalize, covariance matrix, eigenvectors and eigenvalues, sort, project.

8. **Using min-max normalization without checking outliers.** An extreme value can compress most other observations into a small range.

9. **Confusing z-score and min-max.** Z-score is not restricted to [0,1].

10. **Forgetting that discretization can be supervised or unsupervised.** Entropy-based methods use class labels; equal-width and equal-frequency methods do not.

11. **Assuming data cleaning is a one-time activity.** It is iterative.

12. Exam Practice Questions

Try solving these without looking at the explanations above.

Chapter 3 MCQ Test 0/10 answered

1. What are the six dimensions of data quality?

- ○ Accuracy, completeness, consistency, timeliness, believability, interpretability
- ○ Accuracy, speed, security, availability, consistency, storage
- ○ Completeness, speed, compression, reduction, accuracy, cost

2. Which missing-value method uses regression or Bayesian inference?

- ○ Global constant
- ○ Most probable value
- ○ Ignore the tuple

3. What is the main difference between equal-width and equal-frequency binning?

- ○ Equal-width has equal counts
- ○ Equal-frequency has equal numerical widths
- ○ Equal-width has equal numerical ranges; equal-frequency has approximately equal counts

4. Which test is used to detect association between nominal attributes?

- ○ Chi-squared test
- ○ Pearson correlation
- ○ Z-score

5. What is the range of Pearson's correlation coefficient?

- ○ 0 to 100
- ○ -1 to +1
- ○ -∞ to +∞

6. Which technique reduces the number of attributes?

- ◯ Sampling
- ◯ PCA
- ◯ Data cube aggregation

7. Which sampling method is especially useful for imbalanced classes?

- ◯ Stratified sampling
- ◯ SRSWOR only
- ◯ Cluster sampling only

8. Which normalization method uses the mean and standard deviation?

- ◯ Min-max
- ◯ Decimal scaling
- ◯ Z-score

9. Which discretization method uses class labels and entropy?

- ◯ Equal-width
- ◯ Entropy-based
- ◯ Equal-frequency

10. Which compression method allows exact reconstruction?

- ◯ Lossy
- ◯ Lossless
- ◯ PCA

Submit answers

Numerical practice questions

These are additional practice problems based on the methods in your lecture materials.

Q1. Binning

Given the sorted dataset:

$$4, 10, 11, 15, 21, 24, 25, 31, 36$$

Divide into three equal-frequency bins and smooth using:

- Bin mean
- Bin boundaries

Q2. Chi-squared

A dataset contains 100 customers.

- 40 smokers, of whom 30 bought insurance.
- 60 non-smokers, of whom 20 bought insurance.

Construct the contingency table, calculate expected frequencies, calculate χ², and test independence at $\alpha = 0.001$.

Q3. Min-max normalization

Normalize 600 when:

$$Min = 200, \quad Max = 1000$$

Target range:

$$[0, 1]$$

Q4. Z-score normalization

Given:

$$v = 600, \quad Mean = 500, \quad SD = 200$$

Calculate the z-score.

Q5. Decimal scaling

Given:

$$-850,\ 450,\ 920$$

Find j and normalize all values.

Q6. Entropy

A dataset contains:

- 8 Pass
- 2 Fail

Calculate entropy.

Q7. Haar wavelet

Calculate the Haar coefficients for:

$$X = (4, 6, 10, 12)$$

Use:

$$Average = \frac{a + b}{2}$$

$$Detail = \frac{a - b}{2}$$

The lecture slides provide answers for these practice questions, so you can verify your calculations after attempting them.

---

13. Last-Minute Revision Sheet

Before your exam, make sure you can answer these questions confidently:

Question	What you must remember
Why preprocessing?	Improve quality and suitability of data
Six data-quality dimensions?	Accuracy, completeness, consistency, timeliness, believability, interpretability
Missing values?	Six handling strategies
Noisy data?	Binning, regression, outlier analysis
Binning methods?	Mean, median, boundaries
Data integration?	Merge heterogeneous sources
Entity identification?	Determine whether records refer to the same entity
Chi-squared?	Categorical association test
Pearson correlation?	Numerical linear relationship
Covariance?	Joint variation, unit-dependent
PCA?	Reduce dimensions using principal components
Feature selection?	Select useful original attributes
Sampling?	SRSWOR, SRSWR, cluster, stratified

Numerosity reduction?	Regression, histograms, clustering, sampling, cubes
Compression?	Lossless and lossy
Normalization?	Min-max, z-score, decimal scaling
Discretization?	Convert continuous values into intervals
Entropy?	Measure uncertainty
Information gain?	Reduction in entropy
Concept hierarchy?	Detailed values to higher-level concepts

Most important numerical topics to practice:

1. Binning by mean, median, and boundaries.
2. Chi-squared expected frequencies and test.
3. Pearson correlation and covariance.
4. Min-max, z-score, and decimal scaling.
5. Entropy and information gain.
6. Haar wavelet coefficients.

Study recommendation: Do not just memorize the definitions. Practise the numerical problems until you can solve them without referring to the formulas. In this chapter, binning, chi-squared, normalization, entropy, and sampling comparisons are especially important.